

SAR TARGET DETECTION NETWORK BASED ON SALIENCY-COMBINED SINGLE SHOT MULTI BOX DETECTOR

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ABSTRACT

The Single Shot multi-box Detector (SSD) has been successfully applied in synthetic aperture radar (SAR) target detection. Besides, the saliency information in the saliency map has the ability that strengthening the target of interest and suppressing the clutter. It will help to improve the capability of scene understanding. According to the above, a novel SAR target detection network based on saliency-combined SSD is proposed. The proposed method includes two backbone sub-networks, one taking the SAR images as input for extracting the features, and the saliency map obtained from traditional saliency method is used as the input of the other sub-network to obtain the refined saliency information. Through the fusion module is used in multiple scales to integrate the saliency information and the network feature. Finally, we can get the detection results by the convolutional predictors on the multi-scale integrated feature maps. In addition, we apply the dense connection structure in the two sub-networks to utilize context information. The experimental results based on the miniSAR real data show that the proposed method can achieve a good detection performance.

Index Terms— saliency information, synthetic aperture radar (SAR), target detection, single shot multi-box detector (SSD).

1. INTRODUCTION

Synthetic aperture radar (SAR) has the unique capability of providing images in all-weather operating conditions. In recent years, with the rapid development of SAR technology, the acquisition of high-resolution SAR images is becoming relatively easy, which promotes the SAR target detection methods.¹

The conventional SAR target detection methods mainly focus on the constant false alarm rate (CFAR) method. CFAR is an adaptive detection method that can calculate the detection threshold based on the statistical characteristics of local clutter. According to the statistical characteristics of background clutter, the detection performance can be

improved from two aspects: the selection of clutter distribution model (e.g., CFAR based on generalized gamma distribution [1]) and the design of detector (e.g., CFAR based on superpixel [2]). The CFAR methods can achieve a better detection performance in SAR images with simple scenes. However, when the scenes are complex, The CFAR methods often unable to achieve a good detection performance.

The deep learning methods have achieved great success in the field of target detection due to the powerful feature extraction capabilities of convolution neural networks (CNNs). Generally, these methods can be divided into the following two groups. One is the two-stage detectors, e.g., the Faster RCNN [3]. The other is the one-stage detectors, e.g., Single Shot multi-box Detector (SSD) [5]. Two-stage detectors often have a high detection accuracy while one-stage detectors have the advantage of high speed and simple structures. Inspired by great success in target detection of optical images, the use of CNNs in SAR target detection has also obtained the wide application. Wang et al. [6] proposed a SAR target detection method based on SSD and achieved good detection performance.

However, the existing SAR target detection methods often have lots of false alarms and missing alarms in SAR images with complex scenes. Recently, saliency detection, which is an effective method that imitates the ability of human visual attention mechanism, has received wide attention. The saliency detection allocates more computing resources by highlighting the target of interest and suppressing the clutter, thereby improving the capability of scene understanding. The Itti's method [7], the first saliency detection method, use the Gaussian pyramid to extract multi-scale feature maps and then calculate the difference between the central region and its surrounding region through point-to-point subtraction on two feature maps to generate saliency map. Aiming at using the prior knowledge of the size of targets, Wang et al [8] modified Itti's method and achieved a good detection performance.

According to the above analysis, a novel SAR target detection network based on saliency-combined SSD is proposed. The main contributions are as follows.

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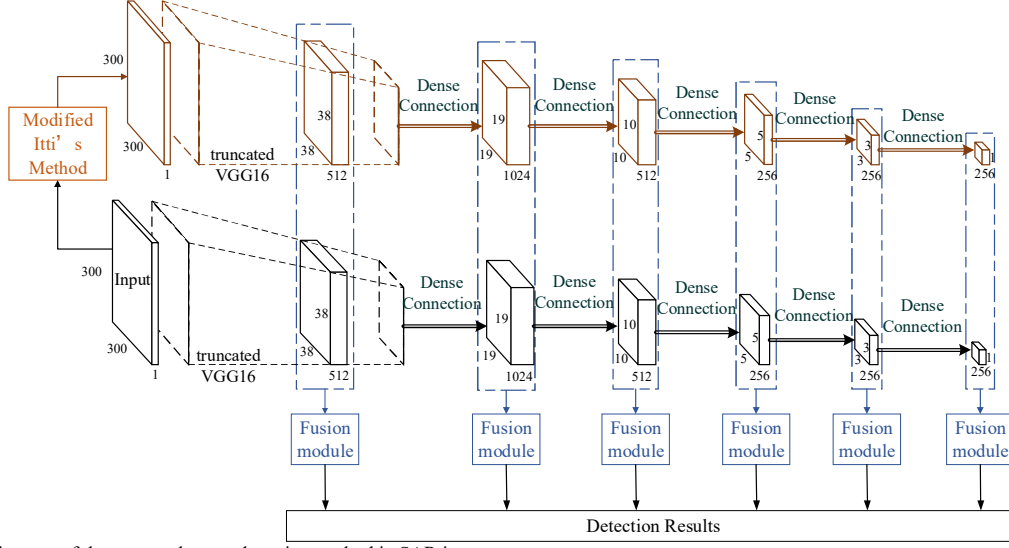


Fig. 1. Architecture of the proposed target detection method in SAR images

- 1) We introduce the saliency information into the SSD network, aiming at highlighting the target of interest and suppressing clutter. It can improve the capability of scene understanding and improve the detection performance.
- 2) The dense connection structure is applied in the proposed saliency-combined SSD method, it can integrate the higher-level feature and the lower-level feature, so as to use the multi-scale feature more effectively and comprehensively.
- 3) We propose a novel multi-level feature fusion module for integrating the saliency information, so as to guide the network strengthens the target of interest. By multiplying the saliency map with the feature map in an element-wise way, it can merge the saliency information into the feature map and improve the detection performance.

Next, we will introduce the proposed method in Section II, give the experimental results and analysis in Section III, and conclude this paper in Section IV.

2. PROPOSED METHOD

Fig.1 shows the architecture of the proposed saliency-combined SSD SAR target detection network. The proposed method includes two separated backbone sub-networks, each of them contains a truncated VGG16 and an auxiliary structure, one taking the original SAR images as input for extracting the features, and the saliency map obtained from the modified Itti's method [8] is used as the input of the other sub-network to obtain the refined saliency information. Please note that the dense connection structure is applied in both two backbone sub-networks, aiming at utilizing the multi-scale information. And then, the fusion module is

applied to integrate the saliency information to guide the network strengthens the target of interest. Finally, we can obtain the detection results after feeding the fused feature maps into the convolutional predictors.

In the following sections, we will introduce the backbone sub-network, the dense connection structure, and the fusion module concretely.

2.1. Backbone sub-networks

We use two same backbone networks in the proposed method. Each of them contains a truncated VGG16 and an auxiliary structure. The architecture of truncated VGG16 is deeper, with a better feature extraction capability. And the auxiliary structure contains some convolutional layers, which is added after truncated VGG16 to further generate feature maps with different resolutions. Therefore, we can obtain multiple layers with different resolutions, which can be used to detect targets of various sizes.

2.2. Dense connection module

The original SSD can detect the target of various sizes due to the structure of multi-level features prediction. However, for each scale, SSD only uses the information of the current feature map. Aiming at fusing the multi-scale features before prediction, we adopt the dense connection module. Fig.2 shows the architecture of the plain structure in the original SSD and the dense connection structure in proposed saliency-combined SSD.

As shown in Fig.2, in the original SSD, the current feature maps $\mathbf{F}_c^{os} \in \mathbb{R}^{m_c \times m_c \times N_c}$ are directly convoluted by $\mathbf{F}_p \in \mathbb{R}^{m_p \times m_p \times N_p}$. Nevertheless, in the dense connection structure, $\mathbf{F}_p \in \mathbb{R}^{m_p \times m_p \times N_p}$ first passes through two different operations respectively. Specifically, on the one hand, a

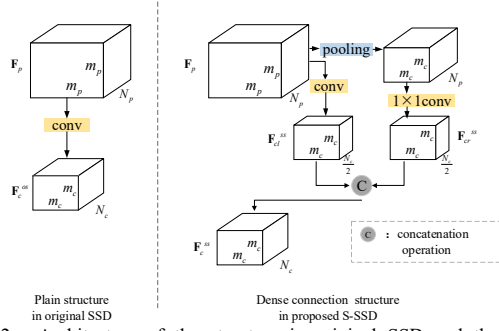


Fig. 2. Architecture of the structure in original SSD and the dense connection structure in proposed method.

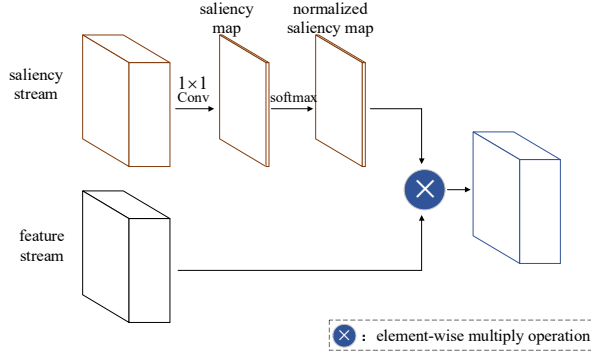


Fig. 3. Architecture of the fusion module.

max-pooling layer and a convolution layer with convolution kernel size of 1×1 are utilized to obtain $\mathbf{F}_{cr}^{ss} \in \mathbb{R}^{m_c \times m_c \times \frac{N_c}{2}}$. On the other hand, a convolution layer with convolution kernel size of 3×3 is utilized to obtain $\mathbf{F}_{cl}^{ss} \in \mathbb{R}^{m_c \times m_c \times \frac{N_c}{2}}$. Finally, we concatenate $\mathbf{F}_{cr}^{ss}$ and $\mathbf{F}_{cl}^{ss}$ as the current feature maps $\mathbf{F}_c^{ss} \in \mathbb{R}^{m_c \times m_c \times N_c}$.

2.3. Fusion module

The fusion module is proposed to integrate the saliency information into the feature maps extracted from the input image.

As shown in Fig.3, we first use a convolution layer with 1×1 filters to generate the saliency map. And then the softmax operation is used to normalize the saliency map:

$$ns_i = \frac{\exp(s_i)}{\sum_{j=1}^n \exp(s_j)}, \quad i \in \{1, 2, \dots, n\} \quad (1)$$

where ns_i represents the saliency score of position i in the normalized saliency map. n represents the number of the spatial position in saliency map. The normalized saliency map and the feature map in the feature stream are dot-multiplied, which can enhance the target area and suppress the clutter.

3. EXPERIMENTAL RESULTS AND ANALYSIS

We validate the proposed method on the miniSAR dataset, in which nine SAR images with the resolution $0.1\text{m} \times 0.1\text{m}$ and the size 1638×2510 pixels for our experiments. Seven images are used for training and two for test. Fig.4 shows the two test SAR images of miniSAR dataset, which we refer to as Image A and Image B respectively.

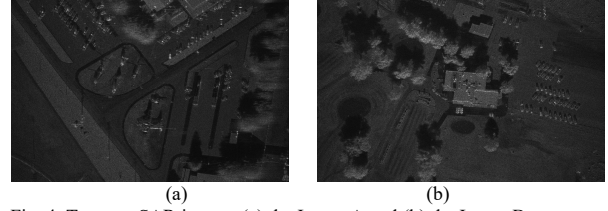


Fig. 4. Two test SAR images (a) the Image A and (b) the Image B.

3.1. Experimental analyses on saliency map

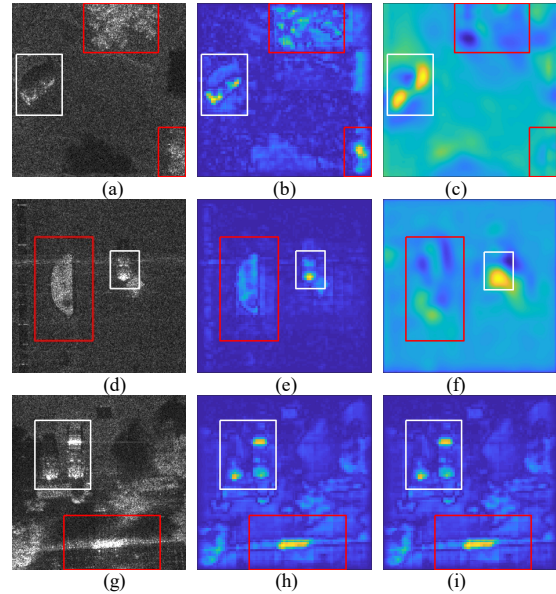


Fig. 5. The sub-images of original big SAR images and their saliency maps obtained by the modified Itti's method and the proposed method respectively. The white rectangles represent targets and the red rectangles represent clutter. (a), (d) and (g) the SAR sub-images, (b), (e), and (h) the saliency maps obtained by the modified Itti's method, (c), (f), and (i) the saliency maps obtained by the proposed method.

In Fig.5, we list the sub-images of original SAR images, their corresponding saliency maps obtained by the modified Itti's method and the proposed method respectively. We can see that the modified Itti's method strengthens some clutter while strengthening the targets of interest. Nevertheless, the proposed method can further learn the saliency information from the saliency map under supervision, it can strengthen the targets of interest while further suppressing the clutter.

3.2. Comparisons with other target detection methods

In order to prove the good detection performance of the proposed method, we compare it with other famous target detection methods, including the conventional Gaussian-CFAR, the Faster R-CNN, and the original SSD. Fig.6 gives the detection results on two test SAR images respectively. In Fig.6, the green rectangles represent the detected target chips, red rectangles represent false alarms and yellow rectangles represent the missing alarms.

As shown in Fig.6, the proposed method has the fewest false alarms and missing alarms compared with other methods. In addition, we count the false alarms and missing alarms and evaluate the overall detection results in terms of precision, recall, and F1-score in Table I. The definitions of those evaluation criteria are as follows:

$$precision = \frac{\text{the number of target chips}}{\text{the total number of chips}} \quad (2)$$

$$recall = \frac{\text{the number of detected targets}}{\text{the total number of targets}} \quad (3)$$

$$F1\text{-score} = \frac{2 \times precision \times recall}{precision + recall} \quad (4)$$

We can conclude that the proposed saliency-combined SSD method, has the highest precision, recall, and F1-score on the test SAR images, which proves that the saliency-combined SSD target detection method is superior to the other above-mentioned target detection methods.

TABLE I EVALUATION OF DIFFERENT TARGET DETECTION METHODS

	Target amounts	Missing alarms	False alarms	precision	recall	F1-score
Gaussian-CFAR	118	24	177	0.3789	0.7966	0.5135
Faster R-CNN		23	23	0.8115	0.8051	0.8083
Original SSD		14	19	0.8468	0.8814	0.8638
Proposed method		4	10	0.9193	0.9661	0.9421

4. CONCLUSION

In this paper, a unified end-to-end detection network, saliency-combined SSD, in which we integrate the saliency information into the SSD network is proposed to achieve SAR target detection. The saliency information can be further learned from the saliency map under supervision, so as to guide the network to strengthen the target area and suppress the clutter. Furthermore, the dense connection module which can integrate the multi-scale information is utilized for better subsequent prediction. The experimental results based on the measured miniSAR dataset show that the proposed method can achieve a better detection performance than other methods.

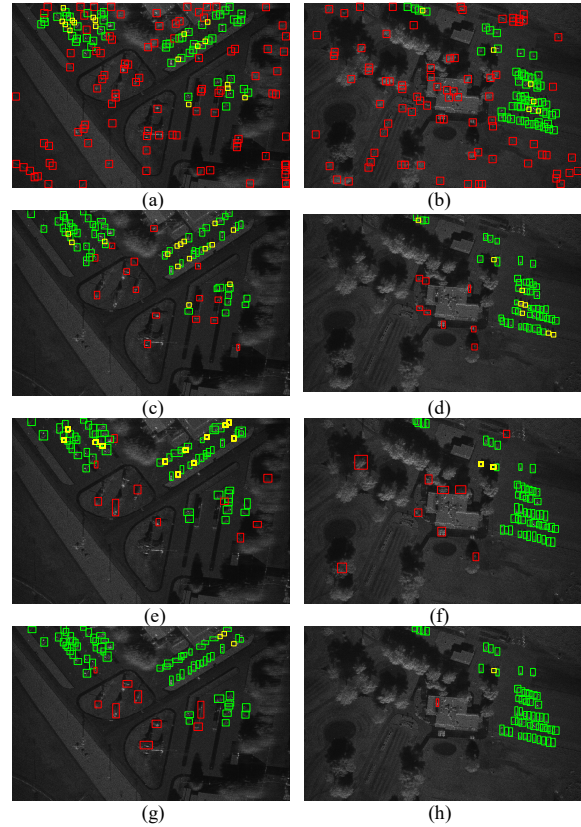


Fig. 6. The detection results comparing with other methods for Image A and B. (a) and (b) the Gaussian-CFAR method, (c) and (d) the Faster R-CNN method, (e) and (f) the original SSD method, (g) and (h) the proposed method, i.e., saliency-combined SSD.

5. REFERENCES

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